

Missing values

$$\vec{X} = (X_1, X_2, \dots, X_p)$$

NA	1	-	'M'
2	NA	-	'S'
NA	3	-	NA
4	5	-	'M'
:	:	-	:
:	:	-	:
:	:	-	:

Let $M = (M_1, \dots, M_p)$ be the NA "mask"

$$\uparrow [X_1 \text{ is NA}]$$

$$\uparrow [X_p \text{ is NA}] = \begin{cases} 1 & X_p \text{ is NA} \\ 0 & X_p \text{ not NA} \end{cases}$$

There are 3 types of missing value mechanisms:

① Missing completely at random (MCAR)

$$P(X_1 \text{ is NA} | X_1, X_2, \dots, X_p) = P(X_1 \text{ is NA}) \Leftrightarrow \{X_1 \text{ is NA}\} \overset{\text{indep.}}{\perp} X_1, \dots, X_p$$

$$P(M=m | \vec{X}) = P(M=m) \quad \uparrow m = (1, 0, \dots, 0)$$

② Missing at random (MAR)

$$P(X_1 \text{ is NA} | X_1, X_2, \dots, X_p) = P(X_1 \text{ is NA} | X_2, \dots, X_p) \Leftrightarrow \{X_1 \text{ is NA}\} \perp X_1 | \{X_2, \dots, X_p\}$$

$$P(M=m | X) = P(M=m | X_{-m})$$

③ Missing not at random (MNAR)

$$P(X_1 \text{ is NA} | X_1, X_2, \dots, X_p) \neq P(X_1 \text{ is NA} | X_2, \dots, X_p) \Leftrightarrow \{X_1 \text{ is NA}\} \not\perp X_1 | \{X_2, \dots, X_p\}$$

$$P(M=m | X) \neq P(M=m | X_{-m})$$

Missing values: Example

X_1	X_2	
1	M	.
5	S	.
NA	M	.
2	S	.
0	M	.
NA	S	.

For each student:

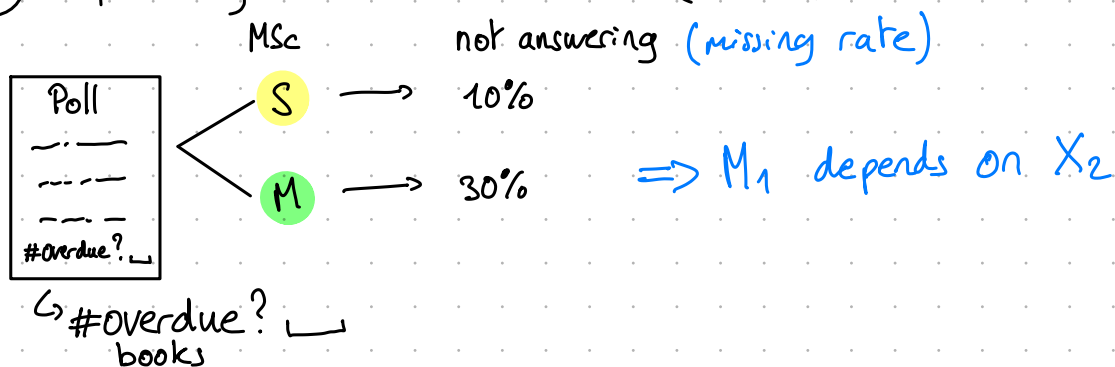
X_1 : #overdue books at the library

X_2 : Management (M)/Statistics (S)

① Missing completely at random (MCAR)

e.g.
 • librarian forgot to type X_1 in,
 • or students did not answer poll,
 randomly 5% of the time
 missing rate (indep. of everything)

② Missing at random (MAR)



③ Missing not at random (MNAR)

e.g. students are uncomfortable answering if #overdue is large:

Poll

#overdue? _

$\hookrightarrow$ #overdue? books

0	1	2	...
10%	15%	30%	

$\rightarrow$ missing rate:

difficult to diagnose

$\Rightarrow M_1$ depends on X_1

Diagnose mechanism

master #overdue

S	0
S	3
S	NA
S	NA
M	2
M	NA
M	1
M	NA

50%

50%

case 1: MCAR

master #overdue

S	0
S	3
S	NA
S	1
M	NA
M	NA
M	1
M	NA

25%

75%

case 2: MAR

Why do we care?

- MCAR: can remove "NA" observations (rows) without introducing bias
- MAR: Removing observations introduces bias

Above, Removing NA observations:

- Case 1: keeps the 50% "S" - 50% "M" distribution
- Case 2: yields a biased 75% "S" - 25% "M" distribution

Solutions?

For single imputation:

① Remove NA observations (rows)

- pros: simple cons: • Only works with MCAR!
↳ introduces bias otherwise
• Might lose information

(Univariate)

② Constant imputation (mean / median / majority category)

- pros: simple cons: • Artificially reduces variability in the data
⇒ might decrease the model's accuracy

(Multivariate)

③ Conditional imputation (lin regr., kNN-imputer, MissForest, ...)

Regress the missing variable using others as covariates
↳ see e.g. `sklearn.impute`

- pros: • considers more info.
• Can often lead to better performance (if features are dependent)
- cons: • computationally more expensive
• Conceptually more involved

④ Remove variable (column)

- pros: • Simple cons: • Might only make sense if most of observations are missing
• Might lose information

use Y values as covariate for conditional imputation?

$\Rightarrow$ Bad idea if you need to predict on new data...

train:

master	#overdue	Y
S	0	.
S	3	.
S	NA	0
S	NA	.
M	2	.
M	NA	.
M	1	.
M	NA	.

Predict Y on new data?

$$X_{\text{train}_1} = (S, 1_{\text{ok}})$$

Y

?

.

$$X_{\text{test}_2} = (M, NA)$$

?

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$\hookrightarrow$ don't have Y value to impute on test if decided to include it during training